

Agent-Based Co-Scientist: Initial Backlog Organized into Epics v0.2

Epic IDs

Old Epic ID	New Epic ID	Epic Name	Story Prefix
E-001	CS-EPIC-GOAL	Applied Research Goal Workspace	CS-GOAL
E-002	CS-EPIC-SCOUT	Research Scout and Evidence Retrieval	CS-SCOUT
E-003	CS-EPIC-ONTOLOGY	Personal Sound Zone Semantic Layer	CS-ONTOLOGY
E-004	CS-EPIC-APPROACH	Approach Card Generation and Curation	CS-APPROACH
E-005	CS-EPIC-SCORE	Evidence-Linked Rubric Scoring	CS-SCORE
E-006	CS-EPIC-HYPOTHESIS	Hypothesis and Combination Generation	CS-HYP
E-007	CS-EPIC-EXPERIMENT	Experiment Card and Spec Generation	CS-EXP
E-008	CS-EPIC-APPROVAL	Human Approval and Execution Handoff	CS-APPROVAL
E-009	CS-EPIC-VALIDATION	Validation Result Ingestion and Score Updates	CS-VALIDATION

E-010	CS-EPIC- DEVICE	Candidate Device Concept Synthesis	CS-DEVICE
E-011	CS-EPIC- ROADMAP	Research Roadmap and Next-Best Experiment Planning	CS-ROADMAP
E-012	CS-EPIC- GOVERNANCE	Co-Scientist Agent Orchestration and Governance	CS-GOV
E-013	CS-EPIC-UI	Co-Scientist User Interface	CS-UI
E-014	CS-EPIC- EVALUATION	Observability, Evaluation, and Quality Metrics	CS-EVAL

User Story IDs

CS-EPIC-GOAL — Applied Research Goal Workspace

New Story ID	Former ID	User Story	Priority
CS-GOAL-001	CS-001	As a researcher, I want to create an applied research goal so that the co- scientist can organize evidence, approaches, experiments, and device concepts	P0

		around a specific objective.	
CS-GOAL-002	CS-002	As a researcher, I want to specify the target application, such as personal sound zones, so that the system can apply the correct ontology and scoring rubric.	P0
CS-GOAL-003	CS-003	As a researcher, I want to define success criteria such as acoustic contrast, dark-zone attenuation, latency, robustness, and calibration burden so that generated experiments are device-relevant.	P0
CS-GOAL-004	CS-004	As a product lead, I want to define device constraints such as speaker count, form factor, compute budget, and acceptable setup time so that	P1

		recommendations reflect practical product limits.	
CS-GOAL-005	CS-005	As a researcher, I want to save and reopen a research goal workspace so that I can continue a multi-session research program.	P1

CS-EPIC-SCOUT — Research Scout and Evidence Retrieval

New Story ID	Former ID	User Story	Priority
CS-SCOUT-001	CS-010	As a researcher, I want the system to retrieve papers, chunks, claims, methods, metrics, and citations relevant to my research goal so that I can start from grounded evidence.	P0
CS-SCOUT-002	CS-011	As a researcher, I want the Research Scout to retrieve evidence for specific method	P0

		families such as acoustic contrast control, pressure matching, beamforming, ANC, and crosstalk cancellation.	
CS-SCOUT-003	CS-012	As a researcher, I want retrieved evidence grouped by method, metric, hardware assumption, and failure mode so that I can compare approaches more easily.	P0
CS-SCOUT-004	CS-013	As a researcher, I want every retrieved evidence item to link back to paper, section, page, chunk, claim, or table so that I can verify the source.	P0
CS-SCOUT-005	CS-014	As a researcher, I want the system to say when evidence is insufficient so that I do not	P0

		mistake speculation for literature support.	
CS-SCOUT-006	CS-015	As a researcher, I want to filter retrieved evidence by year, venue, author, paper collection, method, metric, or application so that I can focus the research scope.	P1

CS-EPIC-ONTOLOGY — Personal Sound Zone Semantic Layer

New Story ID	Former ID	User Story	Priority
CS-ONTOLOGY-001	CS-020	As a researcher, I want the system to maintain a personal sound zone ontology so that related terms and methods are normalized.	P0
CS-ONTOLOGY-002	CS-021	As a researcher, I want the ontology to include acoustic goals such as	P0

		bright-zone fidelity, dark-zone attenuation, acoustic contrast, speech privacy, and robustness to listener movement.	
CS-ONTOLOGY-003	CS-022	As a researcher, I want the ontology to include control methods such as acoustic contrast control, pressure matching, beamforming, null steering, ANC, adaptive filtering, and crosstalk cancellation.	P0
CS-ONTOLOGY-004	CS-023	As a device engineer, I want the ontology to include hardware concepts such as speaker arrays, calibration microphones, feedback microphones, head tracking	P0

		sensors, DSP, and amplifiers.	
CS-ONTOLOGY-005	CS-024	As a researcher, I want the ontology to include physical scene assumptions such as room geometry, reverberation, listener position, source placement, and dark-zone placement.	P1
CS-ONTOLOGY-006	CS-025	As a knowledge manager, I want to review, merge, and edit ontology terms so that duplicate terms and aliases do not pollute the graph.	P1

CS-EPIC-APPROACH — Approach Card Generation and Curation

New Story ID	Former ID	User Story	Priority
CS-APPROACH-001	CS-030	As a researcher, I want the system to generate Approach Cards	P0

		from evidence packs so that I can compare candidate research approaches.	
CS-APPROACH-002	CS-031	As a researcher, I want each Approach Card to include method family, mechanism summary, assumptions, metrics, hardware requirements, risks, and evidence links.	P0
CS-APPROACH-003	CS-032	As a researcher, I want each Approach Card to distinguish direct evidence from inferred synthesis so that I know how much trust to place in the card.	P0
CS-APPROACH-004	CS-033	As a researcher, I want to manually edit, approve, reject, or merge Approach Cards so that expert	P0

		review improves system quality.	
CS-APPROACH-005	CS-034	As a researcher, I want Approach Cards to include unresolved questions and next possible experiments so that the card is actionable.	P0
CS-APPROACH-006	CS-035	As a knowledge manager, I want the system to detect duplicate or near-duplicate Approach Cards so that the workspace remains clean.	P1
CS-APPROACH-007	CS-036	As a researcher, I want Approach Cards to move through statuses such as generated, reviewed, scored, experiment-proposed, tested, validated, refuted, or superseded.	P1

New Story ID	Former ID	User Story	Priority
CS-SCORE-001	CS-040	As a researcher, I want each Approach Card scored against a candidate approach rubric so that I can prioritize promising methods.	P0
CS-SCORE-002	CS-041	As a researcher, I want each score dimension to include a rationale, confidence, and evidence links so that the score is explainable.	P0
CS-SCORE-003	CS-042	As a researcher, I want the system to mark scores as low-confidence when evidence is missing or inferred so that weak claims are not over-weighted.	P0
CS-SCORE-004	CS-043	As a device engineer, I want the rubric to include hardware	P0

		feasibility, real-time feasibility, calibration burden, and form-factor relevance so that device constraints affect ranking.	
CS-SCORE-005	CS-044	As a researcher, I want to compare approaches by individual score dimensions instead of only by total score so that I can reason about tradeoffs.	P0
CS-SCORE-006	CS-045	As a product lead, I want to adjust rubric weights so that the system can optimize for fastest prototype, scientific novelty, robustness, or product feasibility.	P1
CS-SCORE-007	CS-046	As a researcher, I want to view approaches in a Pareto-style comparison so	P2

		that high-risk/high-upside methods are not hidden by average scores.	
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CS-EPIC-HYPOTHESIS — Hypothesis and Combination Generation

New Story ID	Former ID	User Story	Priority
CS-HYP-001	CS-050	As a researcher, I want the system to propose combinations of Approach Cards so that I can explore hybrid solutions.	P1
CS-HYP-002	CS-051	As a researcher, I want each generated hypothesis to explain why the approaches may be compatible so that I can evaluate the reasoning.	P1
CS-HYP-003	CS-052	As a researcher, I want each hypothesis to list required assumptions, expected benefits, failure	P1

		modes, and required experiments so that it is testable.	
CS-HYP-004	CS-053	As a researcher, I want the system to avoid combining approaches with incompatible assumptions unless it explicitly flags the conflict.	P1
CS-HYP-005	CS-054	As a product lead, I want the system to generate both conservative and exploratory hypotheses so that we can balance near-term feasibility with invention upside.	P2

CS-EPIC-EXPERIMENT — Experiment Card and Spec Generation

New Story ID	Former ID	User Story	Priority
CS-EXP-001	CS-060	As a research engineer, I want the system to generate Experiment	P0

		Cards from selected Approach Cards so that promising methods can be tested.	
CS-EXP-002	CS-061	As a research engineer, I want each Experiment Card to include objective, hypothesis, baseline, variables, fixed assumptions, metrics, validation criteria, runtime, and artifacts.	P0
CS-EXP-003	CS-062	As a research engineer, I want the system to generate simulation-first experiments for personal sound zone methods so that we can validate ideas before building hardware.	P0
CS-EXP-004	CS-063	As a research engineer, I want generated experiment specs to be	P0

		exportable as YAML or Python config so that they can run in the experimentation environment.	
CS-EXP-005	CS-064	As a researcher, I want experiments to include baselines such as pressure matching, delay- and-sum beamforming, or fixed-filter control so that comparisons are meaningful.	P0
CS-EXP-006	CS-065	As a researcher, I want the system to include parameter sweeps for speaker count, listener movement, frequency band, room condition, and calibration error so that robustness can be evaluated.	P1
CS-EXP-007	CS-066	As a research engineer, I want	P1

		the system to estimate runtime, cost, required compute, and external SDK usage before execution so that I can decide whether to run the experiment.	
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CS-EPIC-APPROVAL — Human Approval and Execution Handoff

New Story ID	Former ID	User Story	Priority
CS-APPROVAL-001	CS-070	As a researcher, I want to review generated experiment specs before execution so that I can correct assumptions.	P0
CS-APPROVAL-002	CS-071	As a researcher, I want to approve, edit, reject, or duplicate an Experiment Card so that I control what gets run.	P0
CS-APPROVAL-003	CS-072	As an administrator, I want experiments requiring Treble,	P0

		GPU, shared compute, or credentials to require explicit approval so that costs and risks are controlled.	
CS-APPROVAL-004	CS-073	As a research engineer, I want approved experiments to be handed off to the experimentation environment so that they can run reproducibly.	P0
CS-APPROVAL-005	CS-074	As a researcher, I want rejected experiments to preserve the reason for rejection so that the co-scientist can improve future proposals.	P1
CS-APPROVAL-006	CS-075	As an administrator, I want an audit log of approval decisions so that experiment execution is accountable.	P1

CS-EPIC-VALIDATION — Validation Result Ingestion and Score Updates

New Story ID	Former ID	User Story	Priority
CS- VALIDATION- 001	CS-080	As a researcher, I want completed experiment results to link back to the Approach Cards they tested so that evidence accumulates over time.	P0
CS- VALIDATION- 002	CS-081	As a researcher, I want validation results to include metric values, pass/fail status, artifacts, plots, logs, and deviations so that I can inspect experiment quality.	P0
CS- VALIDATION- 003	CS-082	As a researcher, I want approach scores to update after validation results are ingested so that recommendation s reflect experimental evidence.	P0

CS- VALIDATION- 004	CS-083	As a researcher, I want failed experiments to preserve logs and partial artifacts so that failures remain informative.	P0
CS- VALIDATION- 005	CS-084	As a researcher, I want the system to distinguish reproduced, partially reproduced, failed, blocked, and superseded results so that research status is clear.	P1
CS- VALIDATION- 006	CS-085	As a product lead, I want validation results to update the research roadmap so that next experiments reflect what was learned.	P1

CS-EPIC-DEVICE — Candidate Device Concept Synthesis

New Story ID	Former ID	User Story	Priority
CS-DEVICE-001	CS-090	As a device engineer, I want	P1

		the system to generate Device Concept Cards from scored Approach Cards so that research becomes applied design direction.	
CS-DEVICE-002	CS-091	As a device engineer, I want each Device Concept Card to include form factor, acoustic architecture, control stack, sensing/calibration strategy, hardware assumptions, compute needs, and risks.	P1
CS-DEVICE-003	CS-092	As a researcher, I want each Device Concept Card to show supporting evidence, Approach Cards, experiments, and validation results so that I can trace why it was proposed.	P1
CS-DEVICE-004	CS-093	As a product lead, I want	P1

		Device Concept Cards to include unresolved risks and next experiments so that the concept becomes a roadmap, not just a summary.	
CS-DEVICE-005	CS-094	As a device engineer, I want to compare multiple device concepts side by side so that I can choose a prototype direction.	P2
CS-DEVICE-006	CS-095	As a product lead, I want to export a Device Concept Card as markdown, JSON, or report format so that it can be shared with stakeholders.	P2

CS-EPIC-ROADMAP — Research Roadmap and Next-Best Experiment Planning

New Story ID	Former ID	User Story	Priority
CS-ROADMAP-001	CS-100	As a research lead, I want the system to	P1

		recommend the next best experiments so that we reduce uncertainty efficiently.	
CS-ROADMAP-002	CS-101	As a researcher, I want roadmap items ranked by information gain, device relevance, cost, and risk so that I can choose practical next steps.	P1
CS-ROADMAP-003	CS-102	As a researcher, I want the system to identify missing evidence for each promising approach so that I know what must be tested.	P1
CS-ROADMAP-004	CS-103	As a product lead, I want separate lanes for conservative, exploratory, and device-prototype experiments so that the research portfolio is balanced.	P2

CS-ROADMAP-005	CS-104	As a researcher, I want completed experiments to automatically update or retire roadmap items so that the roadmap stays current.	P2
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CS-EPIC-GOVERNANCE — Co-Scientist Agent Orchestration and Governance

New Story ID	Former ID	User Story	Priority
CS-GOV-001	CS-110	As a researcher, I want specialized agents for research scouting, semantic extraction, scoring, experiment design, validation interpretation, and device synthesis so that each task is handled appropriately.	P1
CS-GOV-002	CS-111	As an administrator, I want every agent	P1

		action logged so that generated cards, scores, experiment specs, and recommendations are auditable.	
CS-GOV-003	CS-112	As a researcher, I want agents to cite their evidence and mark speculative claims so that outputs remain trustworthy.	P0
CS-GOV-004	CS-113	As an administrator, I want agents to respect corpus permissions and experiment permissions so that private papers and restricted resources are protected.	P0
CS-GOV-005	CS-114	As an administrator, I want agents to require human approval before running external, costly, or credentialed actions so that	P0

		autonomy is bounded.	
CS-GOV-006	CS-115	As a researcher, I want agent outputs to go through critique/revision loops before being finalized so that quality improves.	P2

CS-EPIC-UI — Co-Scientist User Interface

New Story ID	Former ID	User Story	Priority
CS-UI-001	CS-120	As a researcher, I want a co-scientist workspace dashboard so that I can see the current research goal, evidence, cards, experiments, and roadmap.	P0
CS-UI-002	CS-121	As a researcher, I want an Approach Card review interface so that I can inspect, edit, approve, reject, merge, and	P0

		score generated cards.	
CS-UI-003	CS-122	As a researcher, I want a score explanation panel so that I can see why an approach received each score.	P0
CS-UI-004	CS-123	As a research engineer, I want an Experiment Card editor so that I can review and modify generated experiment specs before execution.	P0
CS-UI-005	CS-124	As a researcher, I want a validation result view so that I can inspect metrics, plots, artifacts, and score updates.	P1
CS-UI-006	CS-125	As a device engineer, I want a Device Concept Card view so that I can inspect candidate	P1

		architectures and risks.	
CS-UI-007	CS-126	As a product lead, I want a roadmap view so that I can see prioritized next actions.	P1

CS-EPIC-EVALUATION — Observability, Evaluation, and Quality Metrics

New Story ID	Former ID	User Story	Priority
CS-EVAL-001	CS-130	As a product owner, I want metrics on Approach Card usefulness so that I can evaluate whether extraction is valuable.	P0
CS-EVAL-002	CS-131	As a product owner, I want metrics on evidence grounding and unsupported claim rate so that I can detect hallucination risk.	P0
CS-EVAL-003	CS-132	As a research lead, I want metrics on experiment	P0

		proposal acceptance rate so that I can evaluate experiment generation quality.	
CS-EVAL-004	CS-133	As a platform owner, I want observability across retrieval, extraction, scoring, experiment generation, execution handoff, and result ingestion so that failures are diagnosable.	P1
CS-EVAL-005	CS-134	As a product owner, I want to track research time saved and user satisfaction so that the product's value is measurable.	P1
CS-EVAL-006	CS-135	As a researcher, I want to provide feedback on cards, scores, experiments, and device concepts so that the	P1

		system can improve.	
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Updated MVP Story Cut

Epic	MVP Stories
CS-EPIC-GOAL	CS-GOAL-001, CS-GOAL-002, CS-GOAL-003
CS-EPIC-SCOUT	CS-SCOUT-001, CS-SCOUT-002, CS-SCOUT-003, CS-SCOUT-004, CS-SCOUT-005
CS-EPIC-ONTOLOGY	CS-ONTOLOGY-001, CS-ONTOLOGY-002, CS-ONTOLOGY-003, CS-ONTOLOGY-004
CS-EPIC-APPROACH	CS-APPROACH-001, CS-APPROACH-002, CS-APPROACH-003, CS-APPROACH-004, CS-APPROACH-005
CS-EPIC-SCORE	CS-SCORE-001, CS-SCORE-002, CS-SCORE-003, CS-SCORE-004, CS-SCORE-005
CS-EPIC-EXPERIMENT	CS-EXP-001, CS-EXP-002, CS-EXP-003, CS-EXP-004, CS-EXP-005
CS-EPIC-APPROVAL	CS-APPROVAL-001, CS-APPROVAL-002, CS-APPROVAL-003, CS-APPROVAL-004
CS-EPIC-VALIDATION	CS-VALIDATION-001, CS-VALIDATION-002, CS-VALIDATION-003, CS-VALIDATION-004

CS-EPIC-DEVICE	CS-DEVICE-001, CS-DEVICE-002, CS-DEVICE-003, CS-DEVICE-004
CS-EPIC-GOVERNANCE	CS-GOV-003, CS-GOV-004, CS-GOV-005
CS-EPIC-UI	CS-UI-001, CS-UI-002, CS-UI-003, CS-UI-004
CS-EPIC-EVALUATION	CS-EVAL-001, CS-EVAL-002, CS-EVAL-003

Updated Sprint Breakdown

Sprint 1 — Goal Workspace and Retrieval Hook

Story ID	Priority
CS-GOAL-001	P0
CS-GOAL-002	P0
CS-GOAL-003	P0
CS-SCOUT-001	P0
CS-SCOUT-004	P0
CS-SCOUT-005	P0

Sprint 2 — Semantic Layer and Approach Card Schema

Story ID	Priority
CS-ONTOLOGY-001	P0
CS-ONTOLOGY-002	P0
CS-ONTOLOGY-003	P0
CS-ONTOLOGY-004	P0
CS-APPROACH-001	P0

CS-APPROACH-002	P0
CS-APPROACH-003	P0

Sprint 3 — Approach Card Review and Scoring

Story ID	Priority
CS-APPROACH-004	P0
CS-APPROACH-005	P0
CS-SCORE-001	P0
CS-SCORE-002	P0
CS-SCORE-003	P0
CS-SCORE-004	P0
CS-SCORE-005	P0

Sprint 4 — Experiment Card Generation

Story ID	Priority
CS-EXP-001	P0
CS-EXP-002	P0
CS-EXP-003	P0
CS-EXP-004	P0
CS-EXP-005	P0
CS-APPROVAL-001	P0
CS-APPROVAL-002	P0

Sprint 5 — Experiment Handoff and Result Ingestion

Story ID	Priority
CS-APPROVAL-003	P0

CS-APPROVAL-004	P0
CS-VALIDATION-001	P0
CS-VALIDATION-002	P0
CS-VALIDATION-003	P0
CS-VALIDATION-004	P0

Sprint 6 — Device Concept and MVP Evaluation

Story ID	Priority
CS-DEVICE-001	P1
CS-DEVICE-002	P1
CS-DEVICE-003	P1
CS-DEVICE-004	P1
CS-UI-001	P0
CS-UI-002	P0
CS-UI-003	P0
CS-UI-004	P0
CS-EVAL-001	P0
CS-EVAL-002	P0
CS-EVAL-003	P0

Updated Dependency Map

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1 CS-GOAL
2   -> CS-SCOUT
3     -> CS-ONTOLOGY
4       -> CS-APPROACH
5         -> CS-SCORE
6           -> CS-HYP
7             -> CS-EXP
8               -> CS-APPROVAL
9                 -> CS-VALIDATION
10                  -> CS-DEVICE
11                    -> CS-ROADMAP

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For implementation tracking, I would use the semantic epic IDs as stable identifiers and avoid renumbering them later. New epics can be added without disrupting the backlog, for example:

1	CS-EPIC-HARDWARE
2	CS-EPIC-DSP
3	CS-EPIC-IP
4	CS-EPIC-BENCH
5	CS-EPIC-CALIBRATION
6	